



Figure 2: Training data size needed by LLM-Prop to achieve state-of-the-art (SOTA) results on predicting different properties. For instance, LLM-Prop achieves SOTA results on predicting band gap and volume with just about 90k training data ($\log_{10}90k \approx 4.95$ on the figure) that corresponds to 35k fewer data than what baselines are trained on. We used log of the training data size on the x-axis for clarity. The performance of each model is calculated on validation set.

Table 6: The contribution of each preprocessing strategy on LLM-Prop performance. We compare the baseline (when the input crystal descriptions and the targets are not touched, and with default T5 tokenizer) and to when all strategies are combined together.

Model	Band gap ↓	Volume ↓	Is-gap-direct ↑
LLM-Prop (baseline)	0.256	69.352	0.796
+ modified tokenizer	0.247	78.632	0.785
+ label scaling	0.242	44.515	N/A
+ [CLS] token	0.231	39.520	0.842
+ [NUM] token	0.251	86.090	0.793
+ [ANG] token	0.242	64.965	0.810
- stopwords	0.252	56.593	0.779
LLM-Prop+all	0.229	42.259	0.857

ranging from 5k to 90k and compare its performance with the baselines and the LLM-Prop model trained on the full data. The results in Figure 2 show that LLM-Prop can achieve SOTA results on predicting band gap and volume with just about 90k training data points ($\log_{10}90k \approx 4.95$ on the figure) corresponding to 35k fewer data points compared to what baselines are trained on. Surprisingly, for volume prediction, LLM-Prop can outperform all GNN-based baselines with just 30k training data points ($\log_{10}30k \approx 4.48$ on the figure) corresponding to about 95k fewer data points than what GNN-based baselines are trained on. These results highlight the efficiency and capabilities of LLM-Prop on predicting the properties of crystalline solids compared to the baselines.

5.2 Ablation studies

How does each LLM-Prop design choice improve its performance? Table 6 shows how each technique we use to make LLM-Prop a state-of-the-art crystal property predictor contributes to its performance. We first naively finetune a T5 encoder